Aowen Fan

Curriculum Vitae

Academy of Mathematics and System Sciences
Chinese Academy of Sciences
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Research Interests

Algebraic topology. Homotopy theory and its interaction with arithmetic and algebraic geometry.

Education

2024–Present **M.Sc. in Mathematics**, *Chinese Academy of Sciences*, Advisor: Foling Zou and Jianzhong Pan
Thesis:"The motivic slice sequence of sphere"

2020–2024 **B.Sc. in Mathematics**, *University of Chinese Academy of Sciences*
Thesis:"Equivariant vector bundles and their homotopy fixed points"

Preprints

1. Aowen Fan, "Infinity operad pair, higher distributivity and spectral Tambara functors" arXiv:2611.09115.
2. Aowen Fan and Foling Zou, "On the Kelly monoidal structure of Λ-sequences and unital operads" arXiv:2602.12738.

Conference presentations

2026.8 "Local objects and localization" International Workshop of Algebraic Topology, Chinese Academy of Sciences.

2025.3 "On the Kelly monoidal structure of unital operads" Conference on Geometric topology and their related topics, Tsinghua Sanya International Mathematic Forum.

Invited talks

2026.11 "Cyclotomic t-structure and topological Cartier modules" Two talks at homotopy reading seminar, MCM

2026.9 "The Adams differential on h_j^3" Two talks at Steenrod algebra seminar, Peking University.

2026.5 "$E(n)$-power operations and isogenies" Two talks at homotopy reading seminar, MCM.

2026.2 "The chromatic nullstellensatz" Chromatic winter seminar, online, recordings available.

2026.1 "Spherical Witt vector and tilt adjunction" Chromatic winter seminar, online, recordings available.

2025.12 "Prismatic cohomology via TC^- and Nygaard filtration" Advanced arithmetic geometry course, Chinese Academy of Sciences.

2025.11 "The q-Hodge filtration via solid THH" Two talks at homotopy reading seminar, MCM.

2025.11 "p-complete THH of the Adams summand " eCHT reading seminar, online.

2025.8 "Synthetic homotopy theory and Segal conjecture" Student's forum, Chinese Academy of Sciences.

2025.4 "The $(\infty, 2)$-category of 6-functor formalism" Three talks at homotopy reading seminar, MCM.

2024.10 "Homotopy limits and colimits" Homotopy reading seminar, MCM.

2024.6 "An introduction to equivariant homotopy theory" Student's forum, Chinese Academy of Sciences.

2023.4 "Riemann surfaces" Complex geometry reading seminar, University of Chinese Academy of Sciences.

2022.9 "The cobordism theory" Differential topology seminar, University of Chinese Academy of Sciences.

Research experience

2026.9-12 Homotopy reading seminar at MCM: Beilinson fiber square.

2026.8 IWoAT summer school and conference.
2026.7 Homotopy reading seminar at MCM:Affine derived schemes.
2026.6-9 Steenrod algebra seminar at Peking University.
2026.3-6 Homotopy reading seminar at MCM:Power operations.
2026.1-4 Motivic Computations in K-theory seminar.
2026.1-5 eCHT reading seminar: Group actions on exotic spheres.
2026.1-2 Chromatic winter seminar: Chromatic Nullstellensatz.
2025.9-12 Homotopy reading seminar at MCM:q-de Rham and Habiro cohomology.
2025.9-12 eCHT Kan seminar.
2025.7 IWoAT summer school and conference.
2025.3-6 Homotopy reading seminar at MCM on miscellaneous topics.
2024.9-12 Homotopy reading seminar at MCM on miscellaneous topics.

Teaching

2025.3-5 Teaching assistant, Algebraic Topology II for graduates.
2025.3-6 Teaching assistant, Linear Algebra.
2024.9-12 Teaching assistant, Algebra I for graduates.
2024.9-12 Teaching assistant, Linear Algebra.
2024.3-6 Teaching assistant, Linear Algebra.
2023.9-12 Teaching assistant, Linear Algebra.

Awards and Grants

2026 National Scholarship
2025 Best Student Award in Chinese Academy of Sciences.
2024 Hua Loo-Keng Scholarship.
2021-2023 University Best Student Award

Academic Service

Upcoming The organizer of the Chromatic winter seminar: Topics to be decided.
2026.8 Assistant to organize International Workshop of Algebraic Topology Summer school at Chinese Academy of Sciences.
2026.1 The organizer of the Chromatic winter seminar: Chromatic Nullstellensatz.

References

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